

Phase 4 Novelty Adjudication

This note records the manual review requested for the top novelty spikes in the three phase-4 transfer recipes. It cross-checks the generated packets:

- baseline novelty packet
- heavy novelty packet
- replay-aware novelty packet

The goal is not to re-score novelty mechanically. The goal is to decide whether the strongest lexical code changes look like:

1. true functional adaptation
2. mixed or local adjustment
3. churn or failed adaptation

The decision rule used here is conservative:

- Functional adaptation: the code change is accompanied by a meaningful behavioral shift and a meaningful score or margin improvement.
- Mixed or local adjustment: the change affects behavior or local score, but the evidence for robust functional gain is incomplete or contradictory.
- Churn or failed adaptation: the code changes substantially, but performance is flat, negative, or behaviorally shallow.

Summary Counts

Recipe	Functional adaptation	Mixed or local adjustment	Churn or failed adaptation
rotating_opponents_holdout_endpoint	2/10	2/10	6/10
rotating_plus_nemesis_novelty_replay	5/10	2/10	3/10
rotating_plus_replay_aware_selection	5/10	2/10	3/10

The main pattern is that the heavier and replay-aware recipes show more genuine functional adaptations among their strongest novelty spikes than the baseline does, but the novelty packets still do not justify a claim of broad strategic invention.

Rotating Opponents Baseline

Case	Manual call	Reason
transfer_territory_control / epoch 53 / run_20260507_210350_b	functional adaptation	Large positive score and margin gain inside the same territorial motif, which looks like within-family functional retuning rather than pure churn.
transfer_territory_control / epoch 37 / run_20260507_201051_a	churn or failed adaptation	Large code and descriptor change, but both score and margin worsened.
transfer_resource_collection_denial / epoch 57 / run_20260507_220338_c	churn or failed adaptation	Resource-collection behavior stayed in the same motif and performance fell slightly.
transfer_resource_collection_denial / epoch 3 / run_20260507_234515_e	mixed or local adjustment	Positive local score and margin gain, but descriptor shift stayed small and the behavior motif did not clearly change.
transfer_resource_collection_denial / epoch 2 / run_20260507_225407_d	churn or failed adaptation	Small behavioral movement with worse score and margin.
transfer_territory_control / epoch 11 / run_20260509_002410_i	churn or failed adaptation	Large apparent strategy move from static_guard to claimer, but the result was strongly negative.

Case	Manual call	Reason
transfer_territory_control / epoch 4 / run_20260509_094819_f	mixed or local adjustment	Absolute score rose sharply, but relative margin still fell, so the functional gain is not clean.
transfer_territory_control / epoch 77 / run_20260508_205712_g	functional adaptation	Clear profile shift from static_guard to balanced plus strong margin improvement.
transfer_resource_collection_denial / epoch 12 / run_20260507_234515_e	churn or failed adaptation	Same behavioral family, modest descriptor movement, and worse performance.
transfer_resource_collection_denial / epoch 73 / run_20260507_225407_d	churn or failed adaptation	Profile moved toward interceptor, but the change lost score and margin and does not show useful transfer evidence.

Rotating + Nemesis + Novelty + Replay

Case	Manual call	Reason
transfer_territory_control / epoch 54 / run_20260509_103847_j	functional adaptation	Strong profile shift and very large score and margin gain.
transfer_territory_control / epoch 42 / run_20260507_145614_b	functional adaptation	Clear move from balanced to claimer with a large territorial performance gain.
transfer_territory_control / epoch 87 / run_20260507_145614_b	mixed or local adjustment	Score rose, but margin still dipped slightly and the visible behavior family stayed the same.
transfer_territory_control / epoch 82 / run_20260507_145614_b	functional adaptation	Large move from balanced to static_guard with strong positive payoff.
transfer_pursuit_evasion / epoch 87 / run_20260509_011052_i	churn or failed adaptation	High lexical novelty with no score or margin gain.
transfer_territory_control / epoch 64 / run_20260507_145614_b	functional adaptation	Very large behavioral and payoff change from static_guard to claimer.
transfer_resource_collection_denial / epoch 82 / run_20260509_103847_j	mixed or local adjustment	Mild but positive improvement in the same resource-collection family; useful, but not strong enough to call a major functional change.
transfer_territory_control / epoch 72 / run_20260507_155959_c	functional adaptation	Large shift toward claimer with strong territorial payoff.
transfer_territory_control / epoch 67 / run_20260509_011052_i	churn or failed adaptation	Large code change and descriptor movement, but score and margin both fell.
transfer_resource_collection_denial / epoch 21 / run_20260508_233016_h	churn or failed adaptation	Mostly lexical movement with lower score and margin.

Rotating + Replay-Aware Selection

Case	Manual call	Reason
transfer_territory_control / epoch 49 / run_20260510_005204_d	churn or failed adaptation	Same territorial family with worse score and margin.
transfer_pursuit_evasion / epoch 15 / run_20260510_043943_i	mixed or local adjustment	Real behavioral shift toward static_guard, but no observable payoff change.
transfer_territory_control / epoch 78 / run_20260509_231935_b	functional adaptation	Large move into a stronger territorial regime with major score and margin gains.
transfer_pursuit_evasion / epoch 23 / run_20260510_035354_h	churn or failed adaptation	Lexical change with no score or margin improvement.
transfer_territory_control / epoch 11 / run_20260510_052825_j	functional adaptation	Clear move from static_guard to claimer with strong margin improvement.
transfer_territory_control / epoch 62 / run_20260510_005204_d	functional adaptation	Strong shift from balanced to static_guard plus strong payoff gain.
transfer_territory_control / epoch 84 / run_20260509_231935_b	functional adaptation	Same broad motif, but a real and useful territorial retuning with positive score and margin gain.
transfer_resource_collection_denial / epoch 8 / run_20260510_052825_j	churn or failed adaptation	Profile changed, but the move reduced both score and margin.

Case	Manual call	Reason
transfer_pursuit_evasion / epoch 14 / run_20260510_000504_c	mixed or local adjustment	There is a real pursuit/evasion style change, but no payoff difference in that epoch.
transfer_pursuit_evasion / epoch 42 / run_20260510_035354_h	functional adaptation	Clear move from static_guard to tagger with a strong positive margin shift.

Interpretation

Three conclusions matter for the phase-4 thesis framing.

First, the baseline packet is still dominated by mixed or failed novelty spikes. That means the simpler rotating-opponent recipe does not show a strong case that its biggest lexical changes were reliable functional improvements.

Second, the heavy and replay-aware recipes do show more genuine functional adaptations, but these are concentrated mostly in territory_control and only occasionally in the other environments. The functional gains are therefore real but uneven, not universal.

Third, even the stronger recipes still contain several large code changes that are mixed or negative. That supports a restrained claim: the extra curriculum machinery appears to help produce more useful adaptive adjustments, but the evidence still does not justify saying that the system broadly invents new high-level strategies whenever code novelty spikes.